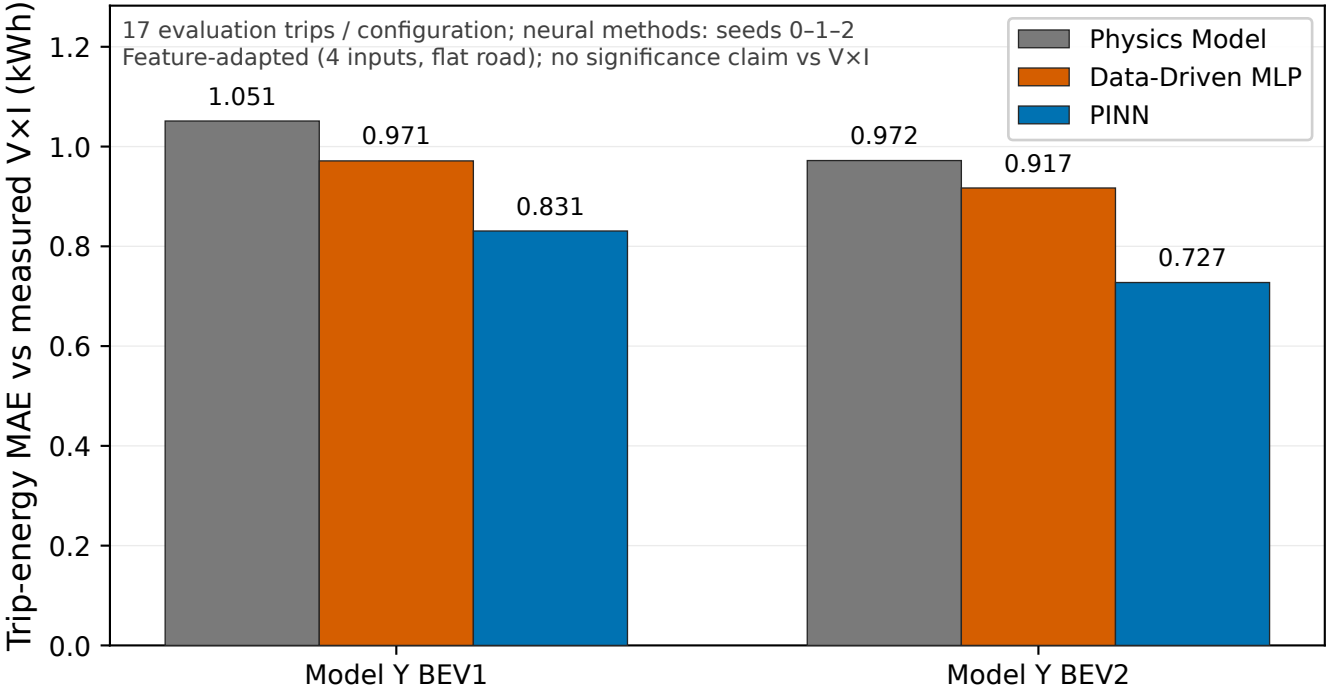
17 evaluation trips / configuration; neural methods: seeds 0–1–2
Feature-adapted (4 inputs, flat road); no significance claim vs V×I
Physics Model
Data-Driven MLP
PINN
Trip-energy MAE vs measured V×I (kWh)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.051
0.971
0.831
0.972
0.917
0.727
Model Y BEV1
Model Y BEV2